

Assignment 1

Q.1)

a) Two types of parallelism in this system are:

1- Task Parallelism

Different grading tasks are executed at the same time. e.g. different servers can grade different student's answer automatically simultaneously.

2- Data Parallelism

The same grading operation is applied to different data at the same time. e.g. multiple processes grading the same question for many students in parallel.

b) Each processor grades a portion of the answer scripts simultaneously instead of waiting for one processor to finish all tasks. e.g. if one server takes 10 min to grade 100 scripts then five servers working in parallel can complete task much faster.

c) One major challenge is synchronization and communication overhead. Processes must communicate results and coordinate with each other, which can cause delays and reduce efficiency if not managed properly.

Q.2)

a) A process is a program that is currently executing in memory. Processes are used because multiple grading tasks must run simultaneously. Each question evaluation can run as a separate process, allowing efficient and organized execution of grading tasks.

b) Four important attributes are :

- 1- Process ID
 - 2- Process State
 - 3- Program Counter
 - 4- Memory Information
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c) When the CPU switches from one process to another, the OS saves the current process information in the PCB and loads the next process information. This allows the system to manage many grading processes efficiently.

Q.3)

a) Two IPC mechanisms could be

- 1- Shared Memory
 - 2- Message Passing
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b) Shared Memory

- Processes share common memory space
- Data can be directly accessed by multiple processes
- Faster communication but requires synchronization

Message Passing

- Processes communicate by sending messages to each other
- No shared memory is required
- Easier to manage but slower due to message transfer

c) Shared memory is better for large volume of data because processes access the same memory region directly without repeatedly copying data through operating system. This reduces communication overhead and improve performance.

Q.4)

a) Four main steps are

1- Create Socket

The server creates a socket using the `socket()` function.

2- Bind the Socket

The socket is assigned an IP address and port number using `bind()`.

3- Listen for connections

The server waits for incoming connection requests using `listen()`.

4- Accept the connection

The server accepts a client connection using `accept()` and establishes communication.

b) After a connection is established, these functions are used for data transmission between client and server.

- `send()`/~~`write()`~~ sends data through the socket.
- `recv()`/~~`read()`~~ receive data from the socket.

Q.5)

a) UDP For live video stream as:

- Low latency is important
- Some packet loss is acceptable
- UDP sends data quickly without waiting for acknowledgment

TCP For large file transfer as:

- Require reliable transmission
 - TCP ensures data is delivered correctly
 - Retransmits lost packets
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b) TCP Characteristics :

- Reliable data delivery
- Error detection & correction
- Guaranteed packet order

TCP Preferred :

- File transfer
- Database synchronisation
- Critical data exchange

UDP Characteristics :

- Faster transmission
- Lower overhead
- No guarantee of delivery or order

UDP Preferred :

- Live video streaming
 - Online gaming
 - Voice communication
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Q.6)

a) The SIMD system with five processing units could provide highest efficiency because :

- SIMD allows all processors to perform the same instruction simultaneously on different data.
 - This maximizes parallel processing and reduces execution time.
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b) No, not always scales linearly.

According to Amdahl's law, some parts of program must run sequentially. As the number of processors increases, the benefit of adding more processes decreases.

Other factors affecting speedup include :

- Communication overhead b/w processors.
 - Synchronization delays
 - Memory access conflicts
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C) Voice data (UDP) = 500 KB/s
Screen-sharing (TCP) = 1000 KB/s

UDP decreases by 40%.

$$= 500 - (40\% \times 500)$$

$$= 500 - 200$$

$$= 300 \text{ KB/s}$$

TCP decreases by 50%.

$$= 1000 - (50\% \times 1000)$$

$$= 1000 - 500$$

$$= 500 \text{ KB/s}$$

$$\text{New transmission rate} = 300 + 500 = 800 \text{ KB/s}$$
